```
Start coding or generate with AI.
```

```
# Hear we import libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# Create a dataframe
df = pd.read_csv('heart.csv')
#print(df)
df.head()
```

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope
0	63	1	3	145	233	1	0	150	0	2.3	0
1	37	1	2	130	250	0	1	187	0	3.5	0
2	41	0	1	130	204	0	0	172	0	1.4	2
3	56	1	1	120	236	0	1	178	0	0.8	2
4	57	0	0	120	354	0	1	163	1	0.6	2

Next steps: Generate code with df   New interactive sheet

```
df.shape
```

```
(303, 14)
```

```
df.columns
```

```
Index(['age', 'sex', 'cp', 'trestbps', 'chol', 'fbs', 'restecg', 'thalach',
       'exang', 'oldpeak', 'slope', 'ca', 'thal', 'target'],
      dtype='object')
```

```
# To get detail info about dataframe
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 303 entries, 0 to 302
Data columns (total 14 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   age       303 non-null    int64
 1   sex       303 non-null    int64
 2   cp        303 non-null    int64
 3   trestbps  303 non-null    int64
```

```
 4   chol       303 non-null    int64
 5   fbs        303 non-null    int64
 6   restecg    303 non-null    int64
 7   thalach    303 non-null    int64
 8   exang      303 non-null    int64
 9   oldpeak    303 non-null    float64
 10  slope      303 non-null    int64
 11  ca         303 non-null    int64
 12  thal       303 non-null    int64
 13  target     303 non-null    int64
dtypes: float64(1), int64(13)
memory usage: 33.3 KB
```

```
# to get statistical information about dataframe
df.describe()
```

	age	sex	cp	trestbps	chol	fbs
count	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000
mean	54.366337	0.683168	0.966997	131.623762	246.264026	0.148515
std	9.082101	0.466011	1.032052	17.538143	51.830751	0.356198
min	29.000000	0.000000	0.000000	94.000000	126.000000	0.000000
25%	47.500000	0.000000	0.000000	120.000000	211.000000	0.000000
50%	55.000000	1.000000	1.000000	130.000000	240.000000	0.000000
75%	61.000000	1.000000	2.000000	140.000000	274.500000	0.000000
max	77.000000	1.000000	3.000000	200.000000	564.000000	1.000000

```
# check missing values
df.isnull().sum()
```

	0
age	0
sex	0
cp	0
trestbps	0
chol	0
fbs	0
restecg	0
thalach	0
exang	0
oldpeak	0
slope	0
ca	0
thal	0
target	0

dtype: int64

```
# check duplicate data items
print(df.duplicated().sum())
```

```
1
```

```
# Drop duplicates
df.drop_duplicates(inplace=True)
df.head()
```

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope
0	63	1	3	145	233	1	0	150	0	2.3	0
1	37	1	2	130	250	0	1	187	0	3.5	0
2	41	0	1	130	204	0	0	172	0	1.4	2
3	56	1	1	120	236	0	1	178	0	0.8	2
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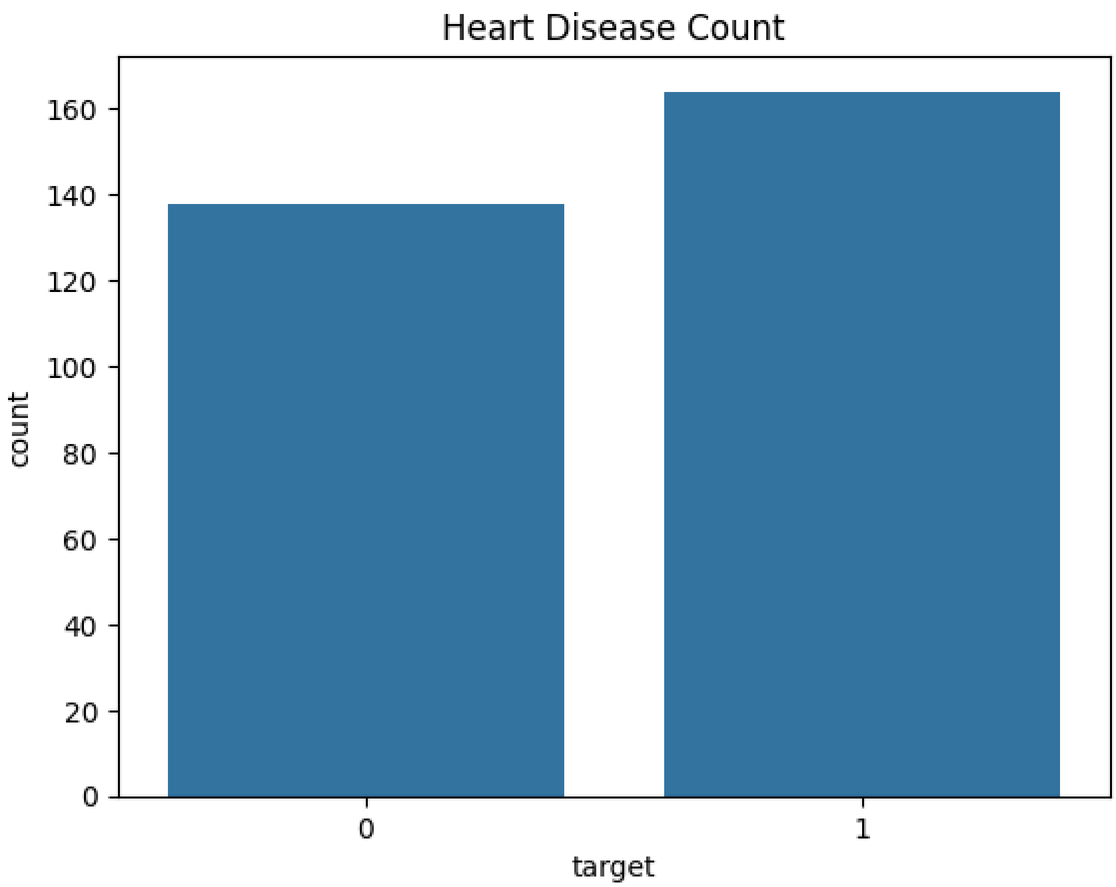
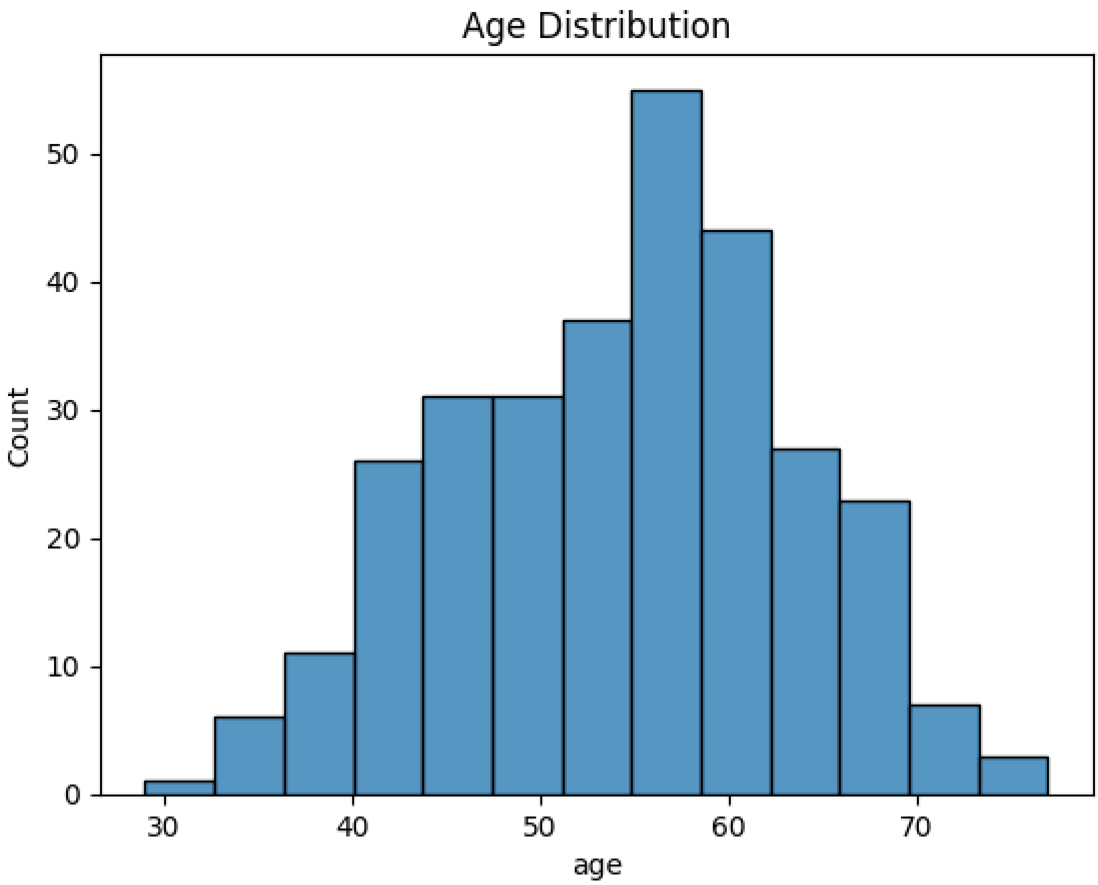
Next steps: Generate code with df New interactive sheet

```
# Data visualization
# Target analysis
```

```
sns.countplot(x='target',data=df)
plt.title('Heart Disease Count')
plt.show()
```



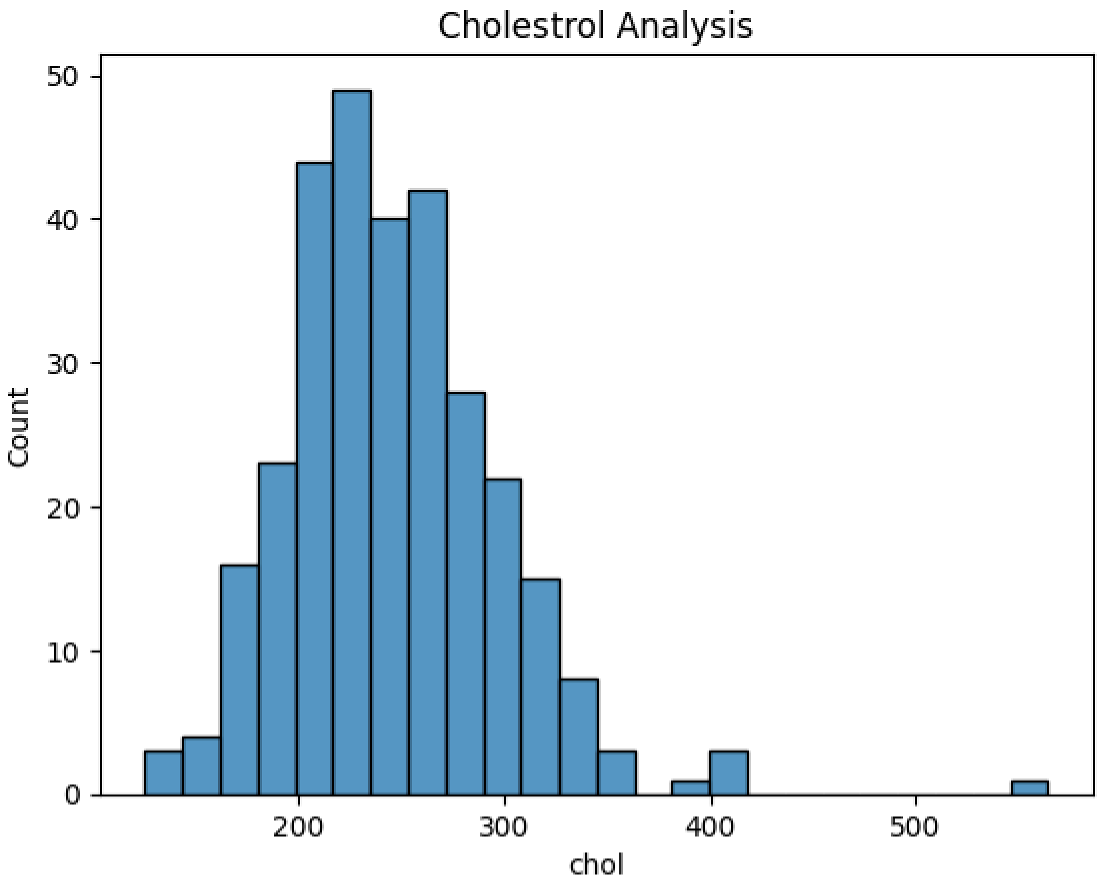

```
# Age Distribution
sns.histplot(df['age'])
plt.title('Age Distribution')
plt.show()
```




```
df.columns
```

```
Index(['age', 'sex', 'cp', 'trestbps', 'chol', 'fbs', 'restecg', 'thalach',
       'exang', 'oldpeak', 'slope', 'ca', 'thal', 'target'],
      dtype='object')
```

```
# Cholestoral Analysis
sns.histplot(df['chol'])
plt.title('Cholestrol Analysis')
plt.show()
```




```
# Gender Analysis
sns.countplot(x='sex', data=df)
plt.title('Gender Analysis')
plt.show()
```

```python
# Correlation Heatmap
plt.figure(figsize=(10,12))
sns.heatmap(df.corr(),annot=True,cmap='coolwarm')
plt.title('Correlation Heatmap')
plt.show()
```